

WEIYUAN DING

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Education

North Carolina State University

Ph.D. in Computer Science (Prospective), Advisor: Prof. Bowen Xu

Aug. 2026 –

Raleigh, NC

North Carolina State University

Master of Computer Science (GPA: 4.0/4.0)

Aug. 2024 – May 2026

Raleigh, NC

Peking University

B.S. in Information and Computing Science

Sept. 2020 – July 2024

Beijing, China

Selected Publications

Smaller = Weaker? Benchmarking Robustness of Quantized LLMs in Code Generation

Mar. 2025

- Study of quantization vs. robustness for code LLMs across *LLaMA*, *DeepSeek*, *CodeGen*, *StarCoder* (350M–33B parameters); quantized models showed higher adversarial/noise robustness (**51.59% vs. 42.86%**).

EVALOOOP: Assessing LLM Robustness in Programming from a Self-consistency Perspective

May 2025

- Loop-based robustness evaluation alternating code generation/summarization; applied to 96 LLMs (e.g., Claude Opus 4, DeepSeek-V3). Observed **2.65%–47.62%** pass@1 drop over 10 loops; robustness not strictly tied to initial accuracy.
- Public leaderboard (100+ models): <https://evaloop.github.io>; under review.

Experience

IDEaS (a SAS Company)

May 2026 – Aug. 2026

Research Intern

Cary, NC

- Prototyping a **RAG-based customer Q&A system** for G3 RMS, IDEaS' hotel revenue management platform: ingestion pipeline turning ~22k heterogeneous internal files into a structured 2,000+-doc corpus; hybrid BM25 + dense retrieval with query expansion, rank fusion, and reranking.
- Designed a two-layer evaluation benchmark (retrieval and generation) with claim-level, expert-verified gold answers from historical support tickets, enabling feasibility assessment without direct ground truth.

Hedra Inc

May 2025 – Jul. 2025

Machine Learning / Data Engineering Intern

San Francisco, CA

- Scaled video data processing throughput from less than 10k/day to **400k clips/day** ($\approx 40\times$) by optimizing I/O (rclone), SLURM parallelism, and module-level compute; built a fully automated raw-to-processed pipeline.
- Built a unified evaluation framework used across model benchmarking and RL training loops; standardized metrics and automated reporting.

North Carolina State University

Feb. 2025 – Dec. 2025

Teaching Assistant / Grader

Raleigh, NC

- **CSC 591: Generative AI for Software Engineering** (Aug. 2025 – Dec. 2025) — Designed rubrics and graded assignments/projects on LLM-based software engineering; authored feedback at scale; supported evaluation scripts.
- **CSC 326: Software Engineering** (Feb. 2025 – May 2025) — Assisted instruction and grading; guided collaborative development best practices.

Baidu (Beijing Baidu Netcom Science Technology Co., Ltd.)

May 2023 – Aug. 2023

Computer Vision Algorithm Intern

Beijing, China

- Standardized data formats and streamlined text-to-image workflows; contributed to data capture/processing pipelines.

Research

AI for Code, Agents, and AI Safety

Nov. 2024 – Present

Graduate Researcher (Advisor: Prof. Bowen Xu), NCSU

Raleigh, NC

- Robustness/evaluation methods for code LLMs (e.g., *EVALOOP*); agentic workflows and safety-oriented benchmarks for programming tasks.

Autonomous Driving Generalized Perception

Jun. 2022 – Sept. 2022

Research Intern (Advisor: Prof. Shanghang Zhang), Peking University

Beijing, China

- Implemented and evaluated BEV perception models (NuScenes); presented analyses in weekly meetings.

Selected Projects

AEGIS: Graph-Guided Multi-Agent for Vulnerability Detection | Core Contributor

2026

- Developed a multi-agent framework utilizing Code Property Graphs (CPG) to decouple structural navigation from semantic analysis, handling complex inter-procedural dependencies within LLM context limits; designed on-demand context retrieval that identifies potential sinks and recursively traces data/control flows.

EVALOOP Leaderboard & Evaluation Suite | Core Contributor

2025

- Engineered pipelines to evaluate 100+ models with reproducible configs; automated ingestion, scoring, and ranking for public site. Code: <https://github.com/evaloop/EvaLoop>

Skills

- Programming: Python, C/C++, Bash, SQL (MySQL)
- ML/LLM: PyTorch, HuggingFace, vLLM, RAG, LLM Agents, LLM Evaluation & Benchmarking
- Systems/Tools: SLURM, Git, GitHub Actions, Vim/VS Code, Linux, tmux, rclone